

Asymptotic Variance in Continuous-Time Markov Chains (CTMC)

Pedagogical Peer Instruction

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1 Introduction and Physical Intuition

In stochastic processes, moving from discrete-time Markov chains (DTMCs) to continuous-time Markov chains (CTMCs) represents a transition from discrete steps to a continuous flow. This transition simplifies the mathematical structure of long-term fluctuations.

To build physical intuition, consider a traveler wandering through a house with several rooms. Each room $x \in S$ has a specific room temperature $f(x)$.

- **Discrete-Time (DTMC):** The traveler teleports to a new room once every minute (discrete steps).
- **Continuous-Time (CTMC):** The traveler stays in each room x for a continuous, exponentially distributed duration of time with rate $q(x)$ before moving to another room.

Over a very long journey, the traveler spends a fraction of their time in room x given by the stationary probability π_x . The long-run average temperature they experience is the **stationary mean**:

$$\bar{f} = \mathbb{E}_\pi[f(X)] = \sum_{x \in S} \pi_x f(x) \quad (1)$$

2 The Value Function and the Poisson Equation

To understand the fluctuations around the stationary mean, we define the **value function** $V(x)$ (also called the expected head-start or deviation function). This function answers the question: *“If the traveler starts in room x today, what is the net cumulative temperature difference they will experience over their entire future journey compared to someone who starts in the long-run average condition?”*

Formally, the value function $V : S \rightarrow \mathbb{R}$ is defined as:

$$V(x) = \int_0^\infty \mathbb{E}[f(X_t) - \bar{f} \mid X_0 = x] dt \quad (2)$$

In a CTMC, the transition rates are governed by the infinitesimal generator matrix Q , where Q_{xy} is the transition rate from state x to state y for $x \neq y$, and $Q_{xx} = -q(x) = -\sum_{y \neq x} Q_{xy}$. By the properties of the generator, the expected rate of change of our future head-start must perfectly balance the instantaneous temperature deviation:

$$QV(x) = -(f(x) - \bar{f}) \quad (3)$$

In vector-matrix form, this is the **continuous-time Poisson equation**:

$$-Q\mathbf{V} = \mathbf{f} - \bar{f}\mathbf{1} \quad (4)$$

Proof of the Poisson Equation. By definition of the generator matrix Q acting on the function V :

$$QV(x) = \lim_{h \rightarrow 0^+} \frac{\mathbb{E}[V(X_h) \mid X_0 = x] - V(x)}{h} \quad (5)$$

Substituting the definition of $V(x)$:

$$\begin{aligned}\mathbb{E}[V(X_h) \mid X_0 = x] &= \int_0^\infty \mathbb{E}[f(X_{t+h}) - \bar{f} \mid X_0 = x] dt \\ &= \int_h^\infty \mathbb{E}[f(X_s) - \bar{f} \mid X_0 = x] ds\end{aligned}\tag{6}$$

Subtracting $V(x)$ from both sides:

$$\mathbb{E}[V(X_h) \mid X_0 = x] - V(x) = - \int_0^h \mathbb{E}[f(X_s) - \bar{f} \mid X_0 = x] ds\tag{7}$$

Dividing by h and taking the limit as $h \rightarrow 0^+$:

$$\begin{aligned}QV(x) &= \lim_{h \rightarrow 0^+} \frac{-1}{h} \int_0^h \mathbb{E}[f(X_s) - \bar{f} \mid X_0 = x] ds \\ &= -(f(x) - \bar{f})\end{aligned}\tag{8}$$

where the last equality follows from the Fundamental Theorem of Calculus since the integrand is continuous. $\square$

3 Asymptotic Variance $\kappa(f)$

If the traveler journeys for a long duration T , the total accumulated temperature they experience is the integral:

$$I_T = \int_0^T f(X_s) ds\tag{9}$$

By the Ergodic Theorem, $I_T/T \rightarrow \bar{f}$ almost surely as $T \rightarrow \infty$. By the Central Limit Theorem (CLT) for continuous-time processes, the scaled fluctuations converge in distribution:

$$\frac{I_T - T\bar{f}}{\sqrt{T}} \implies \mathcal{N}(0, \kappa(f)) \quad \text{as } T \rightarrow \infty\tag{10}$$

The parameter $\kappa(f)$ is the **asymptotic variance** of the reward function on the CTMC:

$$\kappa(f) = \lim_{T \rightarrow \infty} \frac{1}{T} \text{Var} \left(\int_0^T f(X_s) ds \right)\tag{11}$$

Asymptotic variance is different from the stationary variance $\text{Var}_\pi(f(X)) = \mathbb{E}_\pi[(f(X) - \bar{f})^2]$. It captures the temporal correlations (or “stickiness”) of the states. Mathematically, it integrates all autocovariances over time:

$$\kappa(f) = 2 \int_0^\infty \text{Cov}_\pi(f(X_0), f(X_t)) dt\tag{12}$$

4 Derivation of $\kappa(f)$ via the Poisson Equation

Let $Z_t = f(X_t) - \bar{f}$ be the centered reward process. Assuming the process starts in steady state ($X_0 \sim \pi$), we have:

$$\begin{aligned}\text{Var} \left(\int_0^T Z_s ds \right) &= \mathbb{E}_\pi \left[\left(\int_0^T Z_s ds \right)^2 \right] \\ &= \int_0^T \int_0^T \mathbb{E}_\pi[Z_u Z_v] du dv \\ &= 2 \int_0^T \int_0^v \mathbb{E}_\pi[Z_u Z_v] du dv\end{aligned}\tag{13}$$

By stationarity, $\mathbb{E}_\pi[Z_u Z_v] = \mathbb{E}_\pi[Z_0 Z_{|v-u|}]$. Changing variables using $\tau = v - u$:

$$\text{Var} \left(\int_0^T Z_s ds \right) = 2 \int_0^T (T - \tau) \mathbb{E}_\pi[Z_0 Z_\tau] d\tau \quad (14)$$

Dividing by T and taking the limit as $T \rightarrow \infty$ gives the asymptotic variance:

$$\begin{aligned} \kappa(f) &= \lim_{T \rightarrow \infty} \frac{2}{T} \int_0^T (T - \tau) \mathbb{E}_\pi[Z_0 Z_\tau] d\tau \\ &= 2 \int_0^\infty \mathbb{E}_\pi[Z_0 Z_\tau] d\tau \end{aligned} \quad (15)$$

Using the law of total expectation, we can express the integrand as:

$$\mathbb{E}_\pi[Z_0 Z_\tau] = \mathbb{E}_\pi[Z_0 \mathbb{E}[Z_\tau | X_0]] \quad (16)$$

Substituting this back into our expression for $\kappa(f)$:

$$\begin{aligned} \kappa(f) &= 2 \int_0^\infty \mathbb{E}_\pi[(f(X_0) - \bar{f}) \mathbb{E}[f(X_\tau) - \bar{f} | X_0]] d\tau \\ &= 2 \mathbb{E}_\pi \left[(f(X_0) - \bar{f}) \int_0^\infty \mathbb{E}[f(X_\tau) - \bar{f} | X_0] d\tau \right] \end{aligned} \quad (17)$$

Recognizing the inner integral as the value function $V(X_0)$, we obtain the elegant formula:

$$\kappa(f) = 2 \mathbb{E}_\pi[(f(X) - \bar{f})V(X)] \quad (18)$$

5 The Elegant Contrast: DTMC vs. CTMC

To appreciate the simplicity of continuous time, let us compare the asymptotic variance formulas side-by-side:

$$\kappa_{\text{DTMC}}(f) = 2 \mathbb{E}_\pi[(f(X) - \bar{f})V(X)] - \mathbb{E}_\pi[(f(X) - \bar{f})^2] \quad (19)$$

$$\kappa_{\text{CTMC}}(f) = 2 \mathbb{E}_\pi[(f(X) - \bar{f})V(X)] \quad (20)$$

Why does the negative term disappear in CTMC?

In discrete time, when we calculate the variance of a sum of random variables $S_n = \sum_{k=0}^{n-1} Z_k$, the diagonal covariance terms $\text{Cov}(Z_k, Z_k)$ are isolated, discrete point masses. Under steady state, each contributes $\mathbb{E}_\pi[Z_0^2]$. In the double summation, these diagonal terms are counted once, while the off-diagonal terms $\text{Cov}(Z_i, Z_j)$ are counted twice. To relate the total sum to $2 \sum_{k=0}^\infty \mathbb{E}_\pi[Z_0 Z_k]$, we must subtract the single-step variance $\mathbb{E}_\pi[Z_0^2]$ to correct for the double-counting of the diagonal.

In continuous time, there is no discrete “diagonal.” The double integral integrates over a continuous 2D domain $[0, T] \times [0, T]$. The diagonal line $u = v$ has a Lebesgue measure of zero in $\mathbb{R}^2$. Therefore, the instantaneous variance at the points $u = v$ contributes nothing to the continuous integral!

As a result, the continuous-time asymptotic variance is mathematically cleaner, more symmetric, and does not require any discrete correction terms.